



	EE 354/CE 361/MATH 310 - Introduction to Probability and Statistics Fall 2025			
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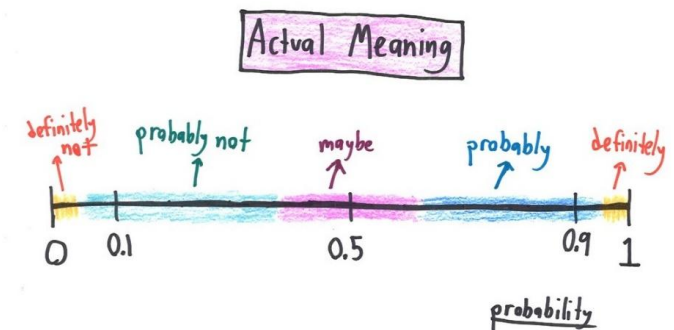


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

Aamir Hasan

Week 10 - Lecture No 19 – October 22, 2025



Announcements!

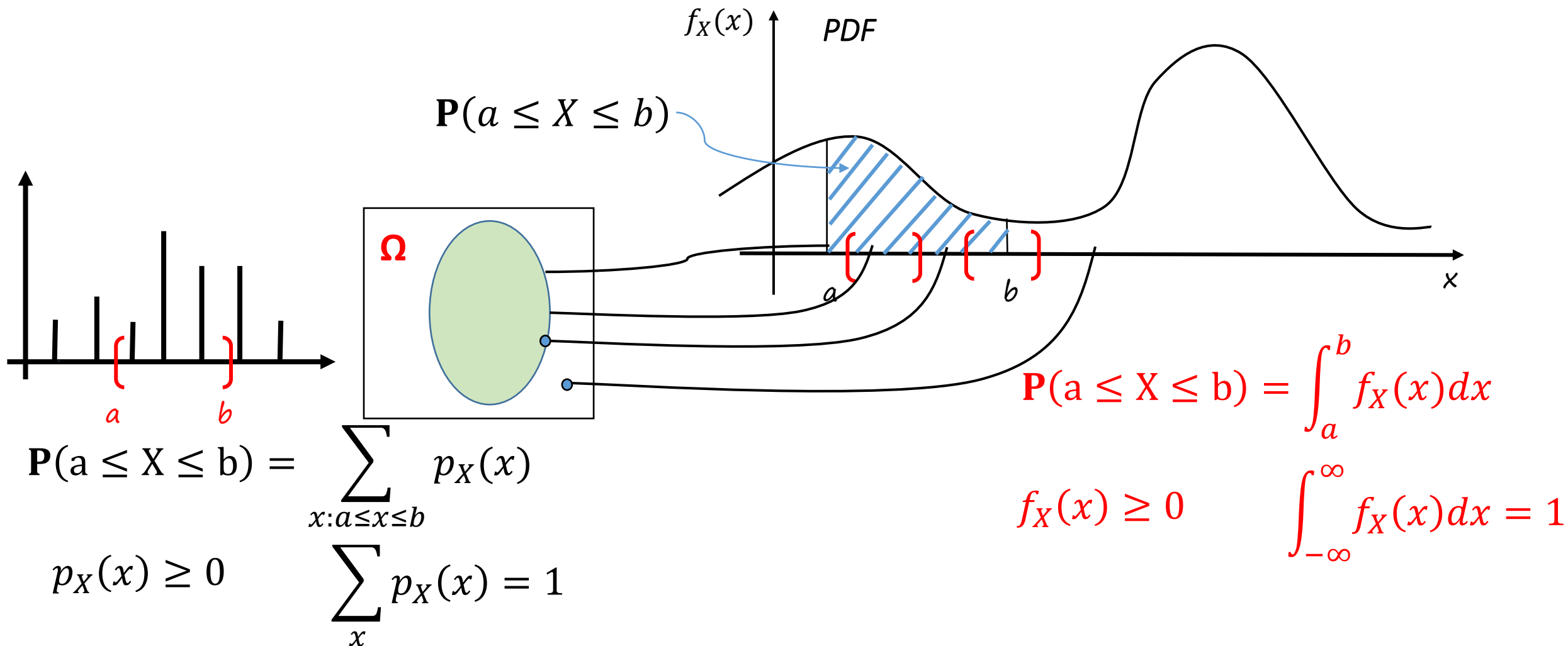
1. All lectures posted.
2. Scores for all 5 quizzes and MT posted.
3. **Quiz No 06** – Monday October 27, 2025 (DRV – Conditional probability and Total Probability)

Agenda for today

1. Continuous Random Variables

Unit 5
Continuous Random Variables

Probability Density Functions (PDFs)



Definition: A random variable is continuous if it can be described by a PDF

$$P(1 \leq X \leq 3 \text{ or } 4 \leq X \leq 5) = P(1 \leq X \leq 3) + P(4 \leq X \leq 5)$$

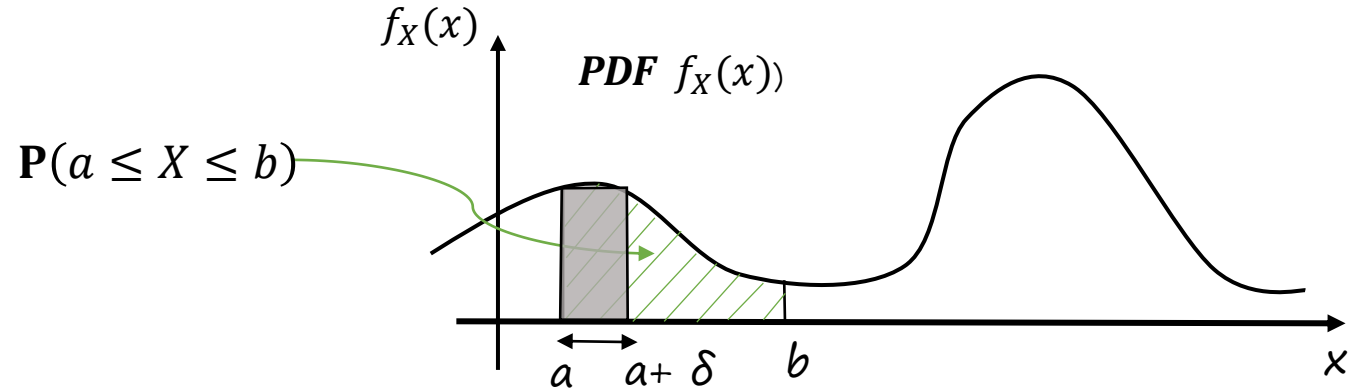
Probability Density Functions (PDFs): contd

$\delta > 0$, small

$$\rightarrow P(a \leq X \leq a + \delta) \approx f_X(a)\delta$$

$$P(a \leq X \leq a + \delta) \approx f_X(a)\delta$$

$$P(X = a) = 0$$



$$P(a \leq X \leq b) = \int_a^b f_X(x) dx$$

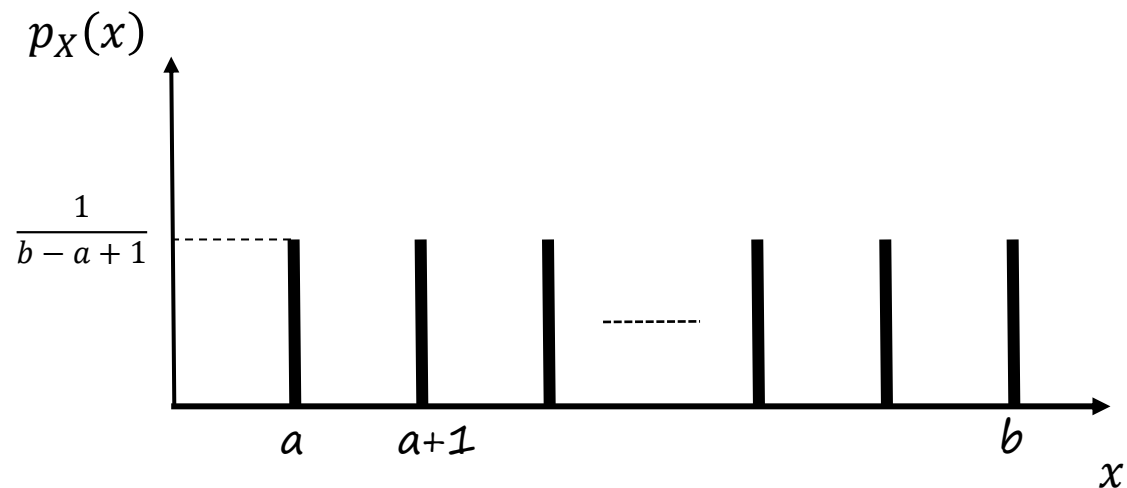
$$f_X(x) \geq 0$$

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

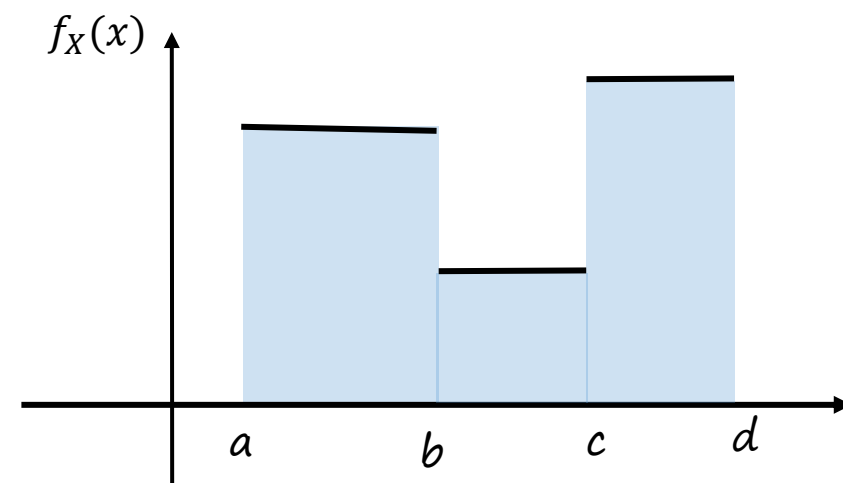
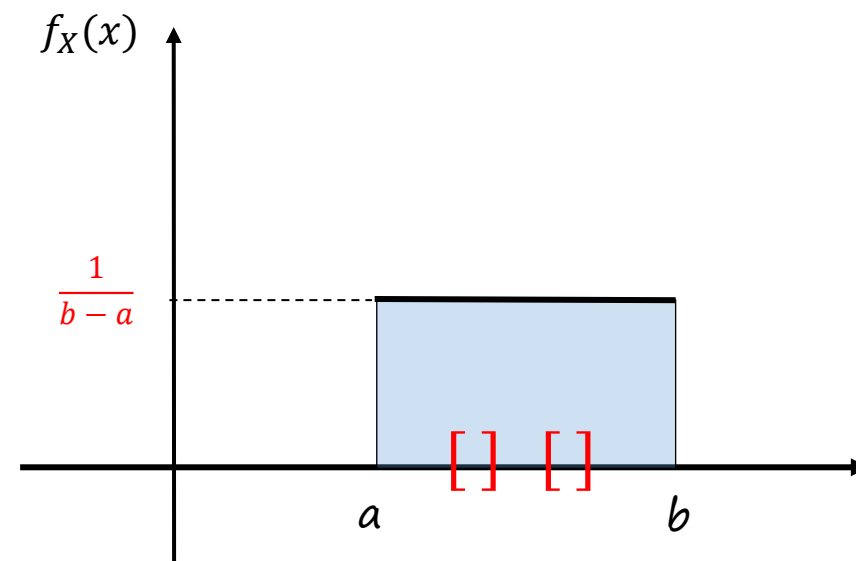
$$\rightarrow P(a \leq X \leq b) = P(\cancel{x = a}) + P(\cancel{x = b}) + P(a < X < b)$$

$$P(a \leq X \leq b) = P(a < X < b)$$

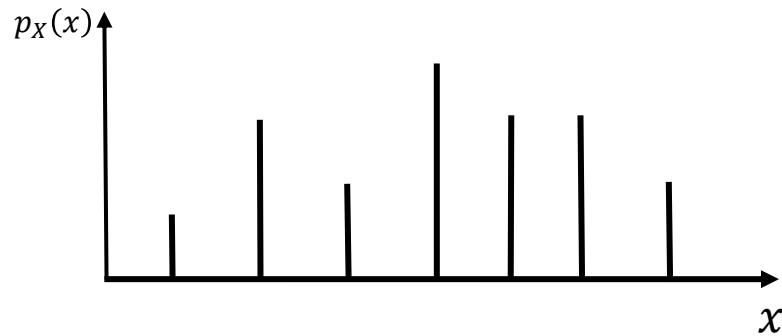
Example: continuous uniform PDF



- Generalization: piecewise constant PDF

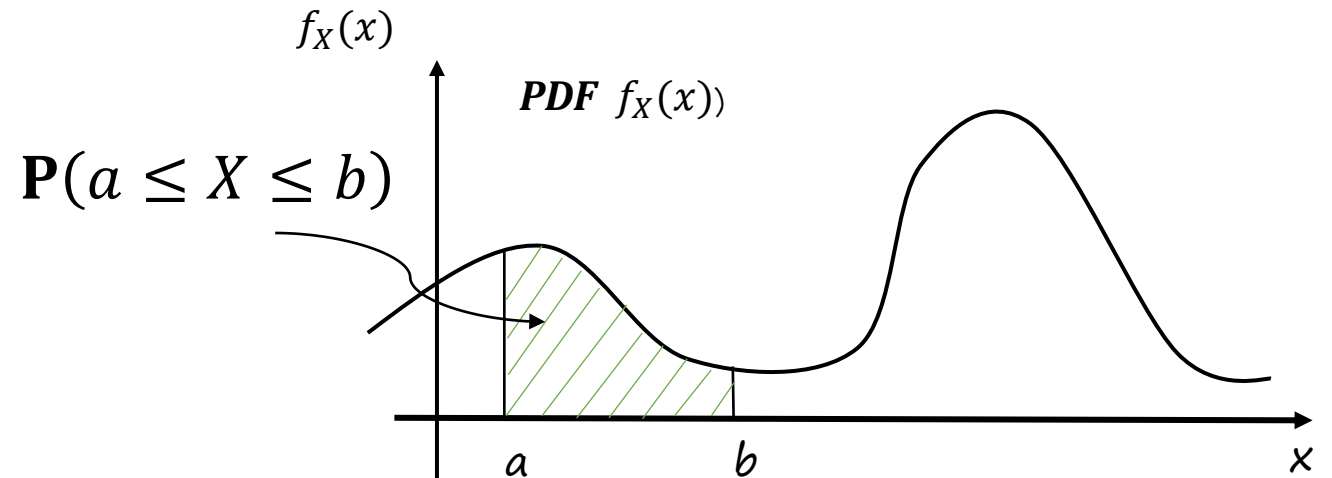


Expectation of a continuous RV



→ $E[X] = \sum_x x p_X(x)$

- **Interpretation:** Average in large number of independent repetitions of the experiment



→ $E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$

Assume, $\int_{-\infty}^{\infty} |x| f_X(x) dx < \infty$